Lab 6: Branch Coverage Report

Name: P T Mounika

SRN: PES1UG24CS311

Repository: L6_L7_06_event_ticket_tier

Files Used

- Application: event_ticket_tier.py
- Test file: BranchCoverageTest_event_ticket_tier.py
- Screenshot: branch_coverage.png

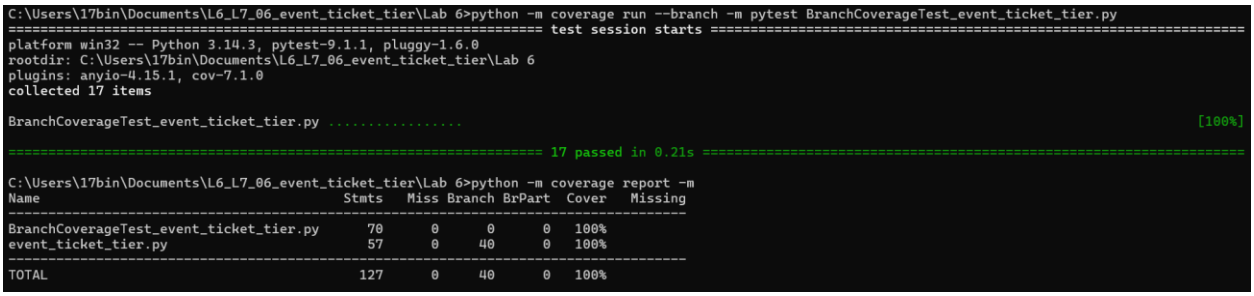
Commands Used

```
python -m coverage run --branch -m pytest BranchCoverageTest_event_ticket_tier.py
python -m coverage report -m
```

Branch Coverage Result

Measure (event_ticket_tier.py)	Result
Statements	57
Missed statements	0
Branches	40
Partially covered branches (BrPart)	0
Coverage	100%

Screenshot



Line Coverage vs Branch Coverage

Line coverage checks whether each line ran at least once. Branch coverage checks whether every decision (if, elif, else, loop) took both its True and False outcomes. Branch coverage is stricter, so it can reveal untested paths even when line coverage is 100%.

Branches Covered

- Input validation: manifests not a list, and negative tax rate
- Manifest confirmed and not confirmed
- Tier selection: vip, premium, general and an invalid tier
- Order quantity valid and quantity less than or equal to 0
- Promo codes: FRAUD_RISK (VIP and non-VIP), EARLYBIRD, and no promo
- Bulk discount: quantity of 5 or more, and fewer than 5
- Exclusive lounge fee: with and without lounge
- Final status: AUDIT_REJECTED, ZERO_SALES and COMPLETED
- monitor_gate_flow: no data, overflow (two high readings in a row), streak reset, and normal flow

Conclusion

The 17 tests written for code coverage also achieved 100% branch coverage, so no extra tests were needed. This is above the required minimum of 90%.